Companion XVIII: Small-Scale Structure in the Relaxation-Driven Cyclic Cosmology Scale-Dependent Growth, Vorticity, and Lyman-α Signatures

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Abstract

Small-scale structure formation provides some of the most sensitive probes of cosmological physics beyond the standard ΛCDM model. Observables such as the persistent S_8 tension, the detailed slope of the Lyman-α forest power spectrum, the surprising abundance of massive galaxies at high redshift, and measurements of cosmic vorticity and velocity dispersion all suggest possible deviations from purely scale-independent growth.

In the Relaxation-Driven Cyclic Cosmology (RDCC), these deviations arise naturally from the bipartite sectoral structure and the universal relaxation mechanism. Sector-dependent sound speeds, different free-streaming lengths, relaxation-induced damping of perturbations, and anisotropic stress generated by inter-sectoral decoherence gradients together produce a characteristic scale-dependent growth function $D(a, k)$.

This Companion derives the complete set of RDCC small-scale structure equations within the Boltzmann framework, computes the resulting matter power spectrum, and identifies clear observational signatures in the Lyman-α forest, weak lensing, redshift-space distortions, and high-redshift galaxy counts. The predicted suppression scale is directly linked to the relaxation rate $\Gamma(a)$, making the mechanism both predictive and testable with existing and upcoming data.

These results establish small-scale structure as a powerful observational window into the fundamental relaxation dynamics of RDCC.

1 From Sectoral Dynamics to Scale-Dependent Growth

In the Relaxation-Driven Cyclic Cosmology (RDCC), the two CPT-conjugate sectors evolve with different temperatures, sound speeds, free-streaming lengths, and relaxation rates. These differences cause the response of matter perturbations to gravitational potentials to become intrinsically scale-dependent. The resulting behaviour is a direct consequence of the bipartite structure introduced in Companion X and the thermodynamic asymmetry derived in Companion XIII.

The key physical ingredients responsible for scale-dependent growth are:

- sector-dependent sound speeds $c_{s,i}(a)$,
- different free-streaming lengths $\lambda_{\mathrm{fs,i}}(a)$,
- relaxation-induced damping of perturbations,

- anisotropic stress generated by inter-sectoral decoherence gradients (Companion XVII),
- a scale-dependent effective gravitational coupling $G_{\text{eff}}(k, a)$.

Together, these mechanisms generate a characteristic scale-dependent growth function $D(a, k)$, a modified matter power spectrum, and non-trivial velocity and vorticity spectra. Importantly, no new particle species or fine-tuned parameters are required; the scale dependence emerges directly from the fundamental relaxation dynamics of RDCC.

1.1 Sectoral Temperatures and Sound Speeds

The two sectors possess different temperatures (Companion XIII),

$$\frac{T_B}{T_A} \approx 0.5, \tag{1}$$

which leads to different sound speeds,

$$c_{s,i}^2 = \frac{\delta p_i}{\delta \rho_i}, \qquad i \in \{A, B\}. \tag{2}$$

Because $c_{s,A} \neq c_{s,B}$, density perturbations respond differently to gravitational potentials in each sector. To leading order,

$$\delta_i(k) \propto \frac{1}{1 + (c_{s,i}^2 k^2/H^2)}, \tag{3}$$

so modes with $k \gg H/c_{s,i}$ are suppressed.

1.2 Sectoral Free-Streaming Lengths

The free-streaming length of each sector is

$$\lambda_{\text{fs,i}}(a) = \int^a \frac{c_{s,i}(a')}{a' H(a')}\, da'. \tag{4}$$

Since $c_{s,A} \neq c_{s,B}$, the two sectors possess different free-streaming scales. Modes with $k > k_{\text{fs,i}} = 1/\lambda_{\text{fs,i}}$ are suppressed in sector i, and the total matter perturbation inherits a weighted combination of these suppressions.

1.3 Relaxation-Induced Damping

The relaxation mechanism introduces additional damping terms in the continuity and Euler equations. The relaxation rate is

$$\Gamma_i(a) = \alpha_i(a) H(a), \tag{5}$$

with $\alpha_A(a) \neq \alpha_B(a)$.

Relaxation suppresses perturbations on scales

$$k \gtrsim k_{\text{rel}}(a) = \frac{\Gamma(a)}{c_{\text{eff}}(a)}, \tag{6}$$

where c_{eff} is the effective sound speed of the combined fluid.

This suppression is smooth, redshift-dependent, and strongest in the Lyman-α regime.

1.4 Decoherence-Gradient Anisotropic Stress

As shown in Companion XVII, inter-sectoral decoherence gradients generate anisotropic stress of the form

$$\pi_{ij}^{(\mathrm{rel})} \propto \partial_{\langle i}\partial_{j\rangle}\Delta_{\mathrm{dec}}, \tag{7}$$

where

$$\Delta_{\mathrm{dec}} = \Gamma_A^{-1} - \Gamma_B^{-1}. \tag{8}$$

This anisotropic stress modifies both scalar and vector perturbations, generating:

- additional suppression of density perturbations near $k \sim k_{\mathrm{rel}}$,
- enhanced vorticity on intermediate scales,
- a modified velocity divergence spectrum.

1.5 Effective Gravitational Coupling

The combined effects of sectoral sound speeds, free-streaming, and relaxation modify the relation between density perturbations and the gravitational potential. The Poisson equation acquires a scale-dependent correction,

$$k^2\Phi(k,a) = 4\pi G_{\mathrm{eff}}(k,a)a^2\rho_{\mathrm{tot}}(a)\,\delta_{\mathrm{tot}}(k,a), \tag{9}$$

with

$$G_{\mathrm{eff}}(k,a) = G\left[1 + \mathcal{O}\left(\frac{k^2}{k_{\mathrm{rel}}^2}\right)\right]. \tag{10}$$

This modification is mild on large scales but becomes significant for $k \gtrsim k_{\mathrm{rel}}$.

1.6 Summary

The bipartite structure and relaxation dynamics of RDCC naturally generate scale-dependent growth through:

- sector-dependent sound speeds and free-streaming lengths,
- relaxation-induced damping of perturbations,
- anisotropic stress from decoherence gradients,
- a scale-dependent effective gravitational coupling.

These ingredients produce a characteristic suppression of small-scale power and enhanced vorticity, providing clear observational signatures in the Lyman-α forest, weak lensing, and redshift-space distortions.

2 RDCC Small-Scale Structure Equations in the Boltzmann Framework

Building directly on the Boltzmann hierarchy developed in Companion XVI, we now derive the perturbed continuity and Euler equations for each sector, including the effects of sector-dependent sound speeds, free-streaming, relaxation-induced damping, and anisotropic stress from inter-sectoral decoherence gradients (Companion XVII). These ingredients combine to produce the characteristic scale-dependent growth function $D(a,k)$.

2.1 Continuity Equation with Relaxation Damping

For each sector $i \in \{A, B\}$, the continuity equation becomes

$$\dot{\delta}_i = -(1 + w_i)(\theta_i - 3\dot{\Phi}) - 3H(c_{s,i}^2 - w_i)\delta_i - \Gamma_i\,\delta_i, \tag{11}$$

where:

- $c_{s,i}^2$ is the sectoral sound speed,
- w_i is the equation of state,
- $\Gamma_i = \alpha_i(a)H(a)$ is the relaxation rate,
- Φ is the Newtonian potential.

The final term represents relaxation-induced damping of density perturbations. It is negligible on large scales but becomes important for $k \gtrsim k_{\rm rel}$.

2.2 Euler Equation with Free-Streaming and Anisotropic Stress

The Euler equation becomes

$$\dot{\theta}_i = -H(1 - 3w_i)\theta_i + \frac{c_{s,i}^2}{1 + w_i}k^2\delta_i + k^2\Psi - \Gamma_i\,\theta_i - k^2\pi_i^{\rm (rel)}, \tag{12}$$

where:

- the $c_{s,i}^2 k^2 \delta_i$ term encodes free-streaming suppression,
- the $-\Gamma_i\theta_i$ term is relaxation-induced momentum damping,
- $\pi_i^{\rm (rel)}$ is the decoherence-gradient anisotropic stress from Companion XVII.

The anisotropic stress term is given by

$$\pi_i^{\rm (rel)}(k) = C_{\rm rel}\,k^2\,\Delta_{\rm dec}(k), \tag{13}$$

with

$$\Delta_{\rm dec} = \Gamma_A^{-1} - \Gamma_B^{-1}. \tag{14}$$

This term enhances vorticity and modifies the velocity divergence spectrum near the relaxation scale.

2.3 Total Matter Perturbation and Effective Growth Equation

The total matter perturbation is

$$\delta_{\rm tot} = \frac{\rho_A\delta_A + \rho_B\delta_B}{\rho_A + \rho_B}. \tag{15}$$

Combining the sectoral continuity and Euler equations yields a modified growth equation for $\delta_{\rm tot}$:

$$\ddot{\delta}_{\rm tot} + [2H + \Gamma_{\rm eff}(a,k)]\,\dot{\delta}_{\rm tot} + \left[c_{\rm eff}^2(a,k)\frac{k^2}{a^2} - 4\pi G_{\rm eff}(a,k)\rho_{\rm tot}\right]\delta_{\rm tot} = S_{\rm rel}(a,k), \tag{16}$$

where:

$$\Gamma_{\text{eff}}(a,k) = \frac{\rho_A\Gamma_A + \rho_B\Gamma_B}{\rho_{\text{tot}}}, \tag{17}$$

$$c_{\text{eff}}^2(a,k) = \frac{\rho_A c_{s,A}^2 + \rho_B c_{s,B}^2}{\rho_{\text{tot}}}, \tag{18}$$

$$G_{\text{eff}}(a,k) = G\left[1 + \mathcal{O}\left(\frac{k^2}{k_{\text{rel}}^2}\right)\right], \tag{19}$$

$$S_{\text{rel}}(a,k) = -k^2\left(\rho_A\pi_A^{(\text{rel})} + \rho_B\pi_B^{(\text{rel})}\right)/\rho_{\text{tot}}. \tag{20}$$

This equation makes explicit that:

- the friction term grows with Γ_{eff},
- the pressure term grows with $c_{\text{eff}}^2 k^2$,
- the gravitational driving term weakens as $G_{\text{eff}}(k)$ decreases,
- the source term S_{rel} enhances vorticity and suppresses growth.

2.4 Definition of the RDCC Growth Function

We define the RDCC growth function $D(a,k)$ via

$$\delta_{\text{tot}}(a,k) = D(a,k)\,\delta_{\text{tot}}(a_{\text{ini}},k). \tag{21}$$

Inserting this into the growth equation yields

$$D'' + \left(\frac{H'}{H} + 2 + \frac{\Gamma_{\text{eff}}}{H}\right)D' + \left[\frac{c_{\text{eff}}^2 k^2}{a^2H^2} - \frac{3}{2}\Omega_m(a)\frac{G_{\text{eff}}(k,a)}{G}\right]D = \frac{S_{\text{rel}}(a,k)}{H^2\,\delta_{\text{tot}}(a_{\text{ini}},k)}. \tag{22}$$

This equation explicitly shows that the growth function becomes scale-dependent whenever:

- $c_{\text{eff}}^2 k^2/H^2$ is non-negligible,
- $\Gamma_{\text{eff}}/H = \mathcal{O}(1)$,
- $G_{\text{eff}}(k,a)$ deviates from G,
- $S_{\text{rel}}(a,k)$ is non-zero.

2.5 Three Growth Regimes

The RDCC growth function exhibits three characteristic regimes:

(i) Large scales ($k \ll k_{\text{fs}}$).

$$D(a,k) \approx D_{\Lambda\text{CDM}}(a). \tag{23}$$

(ii) Intermediate scales ($k \sim k_{\text{fs}}$). Smooth suppression from sectoral sound speeds and free-streaming.

(iii) Small scales ($k \gg k_{\text{rel}}$). Strong suppression from relaxation damping and decoherence-gradient anisotropic stress.

2.6 Summary

The RDCC Boltzmann framework naturally produces scale-dependent growth through:

- sector-dependent sound speeds and free-streaming,
- relaxation-induced damping of density and velocity perturbations,
- anisotropic stress from decoherence gradients,
- a scale-dependent effective gravitational coupling.

These effects combine to yield the RDCC growth function $D(a, k)$ and the characteristic suppression of small-scale power observed in the matter power spectrum.

3 RDCC Predictions for the Matter Power Spectrum

The matter power spectrum in RDCC inherits the full scale dependence of the growth function $D(a, k)$ derived in Section 3. The combined effects of sector-dependent sound speeds, free-streaming, relaxation damping, and decoherence-gradient anisotropic stress produce a smooth, redshift-dependent suppression of small-scale power.

A useful phenomenological approximation is

$$P_{\mathrm{RDCC}}(k, a) = \frac{P_{\Lambda\mathrm{CDM}}(k, a)}{[1 + (k/k_{\mathrm{rel}})^2]^{\nu}}, \tag{24}$$

where:

- $k_{\mathrm{rel}}(a) = \Gamma(a)/c_{\mathrm{eff}}(a)$ is the relaxation scale,
- $\nu \approx 1$–1.5 depends on the sectoral thermodynamics,
- $P_{\Lambda\mathrm{CDM}}$ is the standard matter power spectrum.

This suppression is:

- smooth (unlike the sharp cutoff of warm dark matter),
- redshift-dependent (stronger at high z),
- strongest in the Lyman-α forest regime ($k \sim 1$–$10\, h\,\mathrm{Mpc}^{-1}$),
- mild enough to preserve BAO and large-scale clustering.

Relaxation-induced anisotropic stress additionally enhances vorticity and modifies the velocity divergence spectrum near $k \sim k_{\mathrm{rel}}$, providing further observational handles on the relaxation dynamics.

4 Observational Signatures and Constraints

The scale-dependent growth predicted by RDCC leads to a rich set of observational consequences across weak lensing, redshift-space distortions, the Lyman-α forest, and cosmic web statistics.

4.1 Weak Lensing and the S_8 Tension

Weak lensing surveys measure the combination

$$S_8 = \sigma_8 \sqrt{\Omega_m / 0.3}. \tag{25}$$

RDCC predicts a reduction in σ_8 due to suppressed small-scale growth, naturally alleviating the persistent S_8 tension between CMB and low-redshift probes. The suppression is mild enough to remain consistent with galaxy clustering and BAO.

4.2 Redshift-Space Distortions

Redshift-space distortions probe the scale-dependent growth rate

$$f(a, k) = \frac{d \ln D(a, k)}{d \ln a}. \tag{26}$$

RDCC predicts:

- standard behaviour for $k \ll k_{\rm fs}$,
- mild suppression for $k \sim k_{\rm fs}$,
- strong suppression for $k \gg k_{\rm rel}$.

Future surveys (DESI, Euclid, LSST) will be able to detect or tightly constrain $k_{\rm rel}(z = 0) \sim 0.5$–$2\, h\,{\rm Mpc}^{-1}$.

4.3 Lyman-α Forest

The Lyman-α forest is particularly sensitive to the suppression of power at $k \gtrsim 1\, h\,{\rm Mpc}^{-1}$. RDCC predicts:

- a smooth suppression of the flux power spectrum,
- a redshift-dependent shift in the slope,
- no sharp cutoff (unlike warm dark matter),
- consistency with observed high-redshift galaxy formation.

These features provide a clean observational test of the relaxation scale.

4.4 Vorticity and Velocity Dispersion

Relaxation-induced anisotropic stress (Companion XVII) enhances vorticity on intermediate scales:

$$\omega(k) \propto k^2 \Delta_{\rm dec}(k). \tag{27}$$

This leads to:

- increased velocity dispersion near $k_{\rm rel}$,
- modified cosmic web morphology,
- potential signatures in kinetic SZ and peculiar velocity surveys.

4.5 High-Redshift Galaxy Formation

Because RDCC suppresses only small-scale power, massive halos form similarly to ΛCDM. Thus:

- early massive galaxies remain unaffected,
- JWST observations of high-z galaxies are naturally accommodated,
- only low-mass halos experience suppression.

4.6 Summary of Observational Signatures

RDCC predicts:

- reduced S_8 from weak lensing,
- scale-dependent growth in redshift-space distortions,
- smooth suppression in the Lyman-α forest,
- enhanced vorticity and velocity dispersion,
- preserved BAO and large-scale clustering,
- unaffected formation of massive high-z galaxies.

These signatures provide multiple independent tests of the relaxation dynamics.

5 Conclusion

This Companion has developed a comprehensive framework for small-scale structure formation in the Relaxation-Driven Cyclic Cosmology. Starting from the bipartite sectoral structure and the universal relaxation mechanism, we have shown that RDCC naturally generates scale-dependent growth without introducing new particles or fine-tuned parameters.

The resulting suppression of small-scale power, modified growth function, and enhanced vorticity provide clear, testable predictions across weak lensing, the Lyman-α forest, redshift-space distortions, and cosmic web statistics. The mechanism is tightly linked to the fundamental relaxation rate $\Gamma(a)$, making small-scale structure a powerful observational window into the core dynamics of RDCC.

These results complete the bridge from linear perturbation theory to the quasi-nonlinear regime and set the stage for future detailed studies of halo statistics, galaxy formation, and cosmic web morphology within the relaxation-driven picture.